

COLLEGE ENTRANCE EXAM MATH REVIEWER

Categorized Diagnostic & Practice Examination (UPCAT, PUPCET, USTET Pattern)

Name: _____ Date: _____ Score: _____ / 60

School/Track: _____ Target University: _____

General Instructions: Select the correct answer for each item. This examination is organized sequentially by topic cluster (Items 1 to 60) covering Algebra & Functions, Word Problems, Advanced Algebra & Sequences, Geometry & Analytic Geometry, Trigonometry, Combinatorics & Probability, and Pre-Calculus/Calculus. Work accurately and manage your time effectively.

1. ALGEBRA & FUNCTIONS

1. Simplify: $(2x^3y^{-2})^3 / (4x^{-1}y^4)$
A) $2x^{10} / y^{10}$ B) $x^{10} / (2y^{10})$ C) $2x^8 / y^2$ D) $x^8 / (2y^2)$
2. What are the roots of the quadratic equation $2x^2 - 5x - 3 = 0$?
A) $x = 3, -1/2$ B) $x = -3, 1/2$ C) $x = 3, 1/2$ D) $x = -3, -1/2$
3. Find the value of $\log_2(32) + \log_3(1/81)$.
A) 1 B) 9 C) -1 D) 2
4. Solve for x in the inequality: $|2x - 5| \leq 7$.
A) $-1 \leq x \leq 6$ B) $x \leq -1$ or $x \geq 6$ C) $-6 \leq x \leq 1$ D) $1 \leq x \leq 6$
5. If $f(x) = 3x - 2$ and $g(x) = x^2 + 1$, find $(f \circ g)(3)$.
A) 28 B) 50 C) 10 D) 32
6. Find the inverse function $f^{-1}(x)$ of $f(x) = (2x + 1)/(x - 3)$.
A) $(3x + 1)/(x - 2)$ B) $(x - 3)/(2x + 1)$ C) $(3x - 1)/(x + 2)$ D) $(2x - 1)/(x + 3)$
7. What is the sum of the roots of $3x^2 - 12x + 5 = 0$?
A) -4 B) 4 C) 5/3 D) -5/3
8. If $x - 2$ is a factor of $P(x) = x^3 - kx^2 + 4x - 8$, find k .
A) 0 B) 2 C) 4 D) -2
9. Solve for x : $9^{x+1} = 27^{x-1}$.
A) 3 B) 5 C) 2 D) 4
10. Find the domain of the function $f(x) = \sqrt{16 - x^2}$.
A) $(-\infty, 4]$ B) $[-4, 4]$ C) $[0, 4]$ D) $(-\infty, -\infty)$

2. WORD PROBLEMS

11. A worker can finish a task in 6 hours. Another worker finishes it in 3 hours. How long will it take them working together?
A) 2 hours B) 4.5 hours C) 1.5 hours D) 2.5 hours
12. A solution of 40 liters contains 15% acid. How much water must be added to dilute it to a 10% acid solution?
A) 10 liters B) 15 liters C) 20 liters D) 25 liters
13. The sum of three consecutive integers is 87. What is the largest integer?
A) 28 B) 29 C) 30 D) 31

3. ADVANCED ALGEBRA & SEQUENCES

14. Find the vertical asymptote of $f(x) = (x + 2)/(x^2 - 9)$.

- A) $x = 3$ only B) $x = -3$ only C) $x = 3$ and $x = -3$ D) $y = 0$

15. If $a/b = 3/4$, what is the value of $(2a + b)/(a + 2b)$?

- A) $10/11$ B) $7/11$ C) $5/8$ D) $3/4$

16. What is the remainder when $x^4 - 3x^2 + 5$ is divided by $x + 1$?

- A) 3 B) 7 C) 1 D) -1

17. Find the value of x if $2^x \cdot 4^{x+1} = 64$.

- A) $4/3$ B) 1 C) 2 D) $5/3$

18. Simplify: $(x^2 - 4)/(x^2 - 4x + 4)$.

- A) $(x + 2)/(x - 2)$ B) $(x - 2)/(x + 2)$ C) 1 D) $x/(x - 2)$

19. If $f(x) = x^2 - 3x$, evaluate $(f(x + h) - f(x))/h$.

- A) $2x + h - 3$ B) $2x - 3$ C) $2x + h$ D) $h - 3$

20. What is the 10th term of the arithmetic sequence 3, 7, 11, 15, ...?

- A) 39 B) 43 C) 37 D) 41

21. Find the sum of the infinite geometric series: $12 + 4 + 4/3 + 4/9 + \dots$

- A) 18 B) 16 C) 24 D) 20

4. GEOMETRY & ANALYTIC GEOMETRY

22. What is the distance between points $A(1, 4)$ and $B(7, -4)$?

- A) 8 B) 10 C) 12 D) 14

23. Find the slope of the line perpendicular to $3x - 4y = 12$.

- A) $3/4$ B) $-4/3$ C) $-3/4$ D) $4/3$

24. What is the center and radius of the circle given by $x^2 + y^2 - 6x + 8y = 0$?

- A) Center (3, -4), $r = 5$ B) Center (-3, 4), $r = 5$
C) Center (3, -4), $r = 25$ D) Center (6, -8), $r = 5$

25. The area of a circle is 36π . What is its circumference?

- A) 6π B) 12π C) 18π D) 24π

26. Find the sum of interior angles of a regular octagon.

- A) 1080° B) 1440° C) 900° D) 1260°

27. A right triangle has legs of length 6 cm and 8 cm. What is the length of the altitude drawn to the hypotenuse?

- A) 4.8 cm B) 5.0 cm C) 2.4 cm D) 3.6 cm

28. What is the total surface area of a cube whose volume is 125 cm^3 ?

- A) 150 cm^2 B) 125 cm^2 C) 100 cm^2 D) 180 cm^2

29. Find the equation of the line passing through (2, -1) with a slope of 3.

- A) $y = 3x - 7$ B) $y = 3x - 5$ C) $y = 3x + 7$ D) $y = 3x + 1$

30. Two similar triangles have perimeters of 12 cm and 18 cm. If the area of the smaller triangle is 16 cm^2 , what is the area of the larger triangle?

- A) 24 cm^2 B) 36 cm^2 C) 32 cm^2 D) 48 cm^2

31. What is the volume of a sphere with radius 3 cm?

- A) $36\pi \text{ cm}^3$ B) $108\pi \text{ cm}^3$ C) $12\pi \text{ cm}^3$ D) $27\pi \text{ cm}^3$

32. Find the midpoint of the line segment joining $(-3, 8)$ and $(5, -2)$.

- A) $(1, 3)$ B) $(4, 5)$ C) $(1, 5)$ D) $(-1, 3)$

33. If the diameter of a cylinder is doubled and its height is halved, how does its volume change?

- A) Remains the same B) Doubles C) Quadruples D) Halves

34. Find the length of the diagonal of a rectangular prism with dimensions 3 cm, 4 cm, and 12 cm.

- A) 13 cm B) 15 cm C) 19 cm D) 12.5 cm

5. TRIGONOMETRY

35. An angle measures $5\pi/6$ radians. What is its measure in degrees?

- A) 150° B) 120° C) 210° D) 135°

36. If $\sin(\theta) = 3/5$ and θ is in Quadrant II, find $\cos(\theta)$.

- A) $4/5$ B) $-4/5$ C) $-3/4$ D) $3/4$

37. Simplify the trigonometric identity: $\sin^2(\theta) \cdot \csc(\theta) + \cos^2(\theta) \cdot \sec(\theta)$.

- A) $\sin(\theta) + \cos(\theta)$ B) 1 C) $\tan(\theta) + \cot(\theta)$ D) $\sec(\theta) + \csc(\theta)$

38. What is the exact value of $\tan(225^\circ)$?

- A) 1 B) -1 C) $\sqrt{3}$ D) $-\sqrt{3}/3$

39. A ladder 10 meters long leans against a wall, making a 60° angle with the ground. How high up the wall does the ladder reach?

- A) $5\sqrt{3}$ m B) 5 m C) $10\sqrt{3}$ m D) 8 m

40. Which of the following is equivalent to $\cos(2\theta)$?

- A) $1 - 2\sin^2(\theta)$ B) $2\sin(\theta)\cos(\theta)$
C) $\cos^2(\theta) + \sin^2(\theta)$ D) $2\cos(\theta) - 1$

41. Find the amplitude and period of $f(x) = 4\sin(3x)$.

- A) Amp = 4, Period = $2\pi/3$ B) Amp = 3, Period = $2\pi/4$
C) Amp = 4, Period = 3π D) Amp = 8, Period = $\pi/3$

42. In triangle ABC, side $a = 5$, side $b = 7$, and included angle $C = 60^\circ$. Find side c .

- A) $\sqrt{39}$ B) $\sqrt{74}$ C) 6 D) $\sqrt{109}$

43. Evaluate: $\arcsin(-1/2)$ in radians.

- A) $-\pi/6$ B) $-\pi/3$ C) $5\pi/6$ D) $7\pi/6$

6. COMBINATORICS, PROBABILITY, & STATISTICS

44. In how many ways can 5 books be arranged on a shelf?

- A) 120 B) 25 C) 60 D) 24

45. A committee of 3 students is to be chosen from a group of 8. How many different committees can be formed?

- A) 56 B) 336 C) 24 D) 112

46. Two fair dice are rolled. What is the probability that the sum of the numbers is 8?

- A) $5/36$ B) $1/6$ C) $4/36$ D) $7/36$

47. The mean of 5 numbers is 12. If a 6th number is added, the new mean becomes 14. What is the 6th number?

- A) 24 B) 20 C) 26 D) 18

48. Find the median of the data set: 14, 7, 22, 15, 19, 10, 12.

- A) 14 B) 15 C) 12 D) 13

49. A bag contains 4 red balls, 5 blue balls, and 3 green balls. If one ball is drawn at random, what is the probability that it is NOT blue?

- A) $7/12$ B) $5/12$ C) $1/3$ D) $3/4$

50. In how many distinct ways can the letters of the word "PHILIPPINES" be arranged?

- A) 1,108,800 B) 39,916,800 C) 554,400 D) 2,217,600

51. In how many ways can 6 people sit around a circular table?

- A) 120 B) 720 C) 360 D) 24

7. PRE-CALCULUS & CALCULUS

52. Evaluate the limit: $\lim_{x \rightarrow 3} [(x^2 - 9)/(x - 3)]$.

- A) 6 B) 0 C) 3 D) Does not exist

53. Find the derivative $f'(x)$ of $f(x) = 4x^3 - 5x^2 + 2x - 7$.

- A) $12x^2 - 10x + 2$ B) $12x^2 - 5x + 2$
C) $4x^2 - 10x + 2$ D) $12x^2 - 10x$

54. What is the slope of the tangent line to the curve $y = x^2 + 3x$ at $x = 2$?

- A) 7 B) 10 C) 4 D) 5

55. Evaluate the integral: $\int (3x^2 + 2x) dx$.

- A) $x^3 + x^2 + C$ B) $3x^3 + 2x^2 + C$
C) $6x + 2 + C$ D) $x^3 + 2x^2 + C$

56. Evaluate the definite integral: $\int_0^2 (4x - 1) dx$.

- A) 6 B) 8 C) 4 D) 10

57. Evaluate $\lim_{x \rightarrow \infty} [(5x^2 + 3x)/(2x^2 - 1)]$.

- A) $5/2$ B) ∞ C) 0 D) $3/-1$

58. Find the derivative of $f(x) = (2x + 1)^4$.

- A) $8(2x + 1)^3$ B) $4(2x + 1)^3$
C) $2(2x + 1)^3$ D) $8(2x + 1)^4$

59. Find the critical points of $f(x) = x^3 - 3x$.

- A) $x = 1, -1$ B) $x = 0$ only C) $x = 3, -3$ D) $x = \sqrt{3}, -\sqrt{3}$

60. Evaluate $\lim_{x \rightarrow 0} [\sin(4x)/x]$.

- A) 4 B) 1 C) 0 D) Does not exist